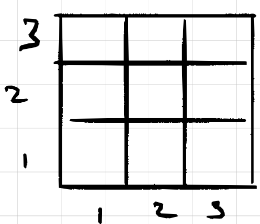
Problem 4.1

Goal: Baye's Rule → warehouse with noisy sensors

Setup: Robot position = (1, 1)
3×3 grid



Define Random variables (considering 2, 1 only)
- D: floor at (2,1) is damaged or not. (D=1) or (D=0)
- C: Detect creaking at (1,1) or not. (C=1) or (C=0)

Prior:
- $P(D=1) = 0.15$ 15% of floor damaged

- Sensor model
  $P(C=1 \mid D=1) = 0.85$ (Given true positive)
  $P(C=1 \mid D=0) = 0.08$ (Given false positive)

1) No creaking observed. Compute posterior probability that (2,1) has damaged floor.

aka what is $P(D=1 \mid C=0)$?

step 1: what are the likelihood of $C=0$

$P(C=0 \mid D=1) = 1 - 0.85 = 0.15$ (false negative)

$P(C=0 \mid D=0) = 1 - 0.08 = 0.92$ (true negative)

step 2: Compute $P(C=0)$ via law of probability

$$P(C=0) = P(C=0 \mid D=1)P(D=1) + P(C=0 \mid D=0)P(D=0)$$
$$= (0.85)(0.15) + (0.92)(1 - 0.15)$$
$$= 0.9095$$

step 3: Apply Baye's Rule

$$P(D=1 \mid C=0) = \frac{P(C=0 \mid D=1)P(D=1)}{P(C=0)}$$
$$= \frac{(0.15)(0.15)}{0.9095}$$
$$\approx 0.0247$$

2) Creaking observed at (1,1). Compute the posterior probability that $P(D=1)$ at (2,1)

aka what is $P(D=1 \mid C=1)$?

step 1: All likelihoods that $C=1$

$P(C=1 \mid D=1) = 0.85$

$P(C=1 \mid D=0) = 0.08$

step 2: Compute $P(C=1)$ via law of Probability

$$P(C=1) = P(C=1 \mid D=1)P(D=1) + P(C=1 \mid D=0)P(D=0)$$

$$= (0.85)(0.15) + (0.08)(1-0.15)$$

$$= 0.1955$$

step 3: Apply Baye's Rule

$$P(D=1 \mid C=1) = \frac{P(C=1 \mid D=1)P(D=1)}{P(C=1)}$$

$$= \frac{(0.85)(0.15)}{0.1955}$$

$$\approx 0.6522$$

3) Sensor Comparison

New sensor sensor model

$P_N(C=1 \mid D=1) = 0.95$ (true positive)

$P_N(C=1 \mid D=0) = 0.02$ (false positive)

3-1) No cracking. Compute $P_N(D=1 \mid C=0)$

Step 1: likelihood of $C=0$

$P_N(C=0 \mid D=1) = 1 - 0.95 = 0.05$

$P_N(C=0 \mid D=0) = 1 - 0.02 = 0.98$

Step 2: Compute $P(C=0)$ via Law of Probability

$$P_N(C=0) = P_N(C=0 \mid D=1)P_N(D=1) + P_N(C=0 \mid D=0)P_N(D=0)$$
$$= (0.05)(0.15) + (0.98)(1-0.15)$$
$$= 0.908$$

Step 3: Apply Baye's Rule

$$P_N(D=1 \mid C=0) = \frac{P_N(C=0 \mid D=1)P_N(D=1)}{P_N(C=0)}$$
$$= \frac{(0.05)(0.15)}{0.908}$$
$$\approx 0.00826$$

Compared to 0.0247, this is a 66.559% improvement

2.1) Creaking at (1,1). Compute posterior probability that $P(D=1)$ at (2,1)

$$P_N(D=1 \mid C=1)?$$

Step 1: All likelihood of $C=1$

$$P_N(C=1 \mid D=1) = 0.95 \quad \text{T.P.}$$
$$P_N(C=1 \mid D=0) = 0.02 \quad \text{F.P.}$$

Step 2: Compute $P_N(C=1)$ via Law of Pobability

$$P_N(C=1) = P_N(C=1 \mid D=1)P_N(D=1) + P_N(C=1 \mid D=0)P_N(D=0)$$
$$=(0.95)(0.15)+(0.02)(1-0.15)$$
$$=0.1595$$

Step 3: Apply Baye's Rule

$$P_N(D=1 \mid C=1) = \frac{P_N(C=1 \mid D=1)\,P(D=1)}{P_N(C=1)}$$
$$= \frac{(0.95)(0.15)}{0.1595}$$
$$\approx 0.8934$$

Compared to 0.6522, this is a 36.985% improvement.

4. Multiple adjacent squares

Define Random variables

D_1 : Damage at (2,1), Prior $P(D_1=1) = 0.15$

D_2 : Damage at (1,2), Prior $P(D_2=1) = 0.15$

C : Creaking at (1,1). $P(C=1)$ or $P(C=0)$

Assume D_1 & D_2 are independent of each other

$P(C=1 \mid \text{at least one damage}) = 0.85$ (given T.P.)

$P(C=1 \mid \text{none damaged}) = 0.08$ (given F.P.)

Step 1 : Compute $P(C=1 \mid \text{at least one damaged})$

$$P(\text{at least one damaged}) = 1 - P(D_1=0)P(D_2=0)$$
$$= 1-(1-0.15)(1-0.15)$$
$$= 0.2775$$

Step 2 : Compute $P(C=1)$ via total probability

$$P(C=1) = P(C=1 \mid \text{at least one})P(\text{at least one}) + P(C=1 \mid \text{none})P(\text{none})$$
$$= (0.85)(0.2775) + (0.08)(1-0.2775)$$
$$\approx 0.294$$

Step 3 : To find $P(D_1=1 \mid C=1)$ we need to condition on all four (D_1, D_2) combinations.
Let A denote the event "At least one damaged"

$$P(D_1=1, C=1) = \sum_{d_2} P(C=1 \mid D_1=1, D_2=d_2)P(D_1=1) \cdot P(D_2=d_2)$$

when $D_1=1$, at least one square is damaged regardless of D_2 so

$P(C=1 \mid D_1=1, D_2=d_2) = 0.85$ for both values of d_2

$$P(D_1=1, C=1) = (0.85)(0.15)(0.15) + (0.85)(0.15)(1-0.15)$$
$$= 0.1275$$

$$\therefore P(D_1=1 \mid C=1) = \frac{P(D_1=1, C=1)}{P(C=1)}$$
$$= \frac{0.1275}{0.294}$$
$$\approx 0.434$$